# PHASE PLANE ANALYSIS FOR THE FITZHUGH-NAGUMO MODEL AND BURSTING

G. Carol, F. Estrella and A. Ruiz-Sebastián

**Abstract:** In this document, we analyse the FitzHugh–Nagumo model to describe the electrical activity of a single neuron. We characterize the different regimes of neuronal activity as a function of the input current by examining the bifurcation diagram of the differential equations defining the model, and we study the system's dynamics within each regime. In the final section, we extend the analysis to a bursting neuron model.

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# 1 PHASE PLANE ANALYSIS

## 1.1 Analysis of the System

Let us consider the FitzHugh-Nagumo 2D dynamical system:

$$\dot{v} = -v(v-1)(v-a) - w + I_{\text{app}}, \tag{1}$$
$$\dot{w} = \varepsilon(v - \gamma w), \tag{2}$$

where:

1. The variable $v \in \mathbb{R}$ is the *membrane potential*.
2. The variable $w \in \mathbb{R}$ is the *gating variable*.
3. The parameters $a = 0.25$, $\gamma = 2.54$, and $\epsilon = 0.01$ are positive constants whose values depend on the specific biological model under consideration.
4. The variable $I_{\text{app}}$ denotes the applied (positive) *current value*, which will be interpreted as an adjustable parameter.

Since we take $0 < \varepsilon \ll 1$, from (2) we have that the gating variable $w$ evolves at a slower rate than the membrane potential $v$. Therefore, these variables are also referred to as the *slow variable* $w$ and the *fast variable* $v$.

Let us consider the nullclines and equilibrium points of the system. First, we note that the $w$-nullcline is the line

$$w = \frac{1}{\gamma} v, \tag{3}$$

whereas the $v$-nullcline is given by the cubic in the $v-w$ plane:

$$w = -v^3 + (a+1)v^2 - av + I_{\text{app}}. \tag{4}$$

We observe that the constant $\gamma$ determines the slope of the $w$-nullcline, $a$ controls the slope and curvature of the cubic and $I_{\text{app}}$ determines its intersection with the $w$-axis.

These parameters also determine the intersections of the nullclines, that is, the equilibrium points of the system. We denote them by

$$(v^*, w^*) = (v^*, \frac{1}{\gamma} v^*), \tag{5}$$

where $v^*$ is a solution of the cubic equation:

$$v^3 - (a+1)v^2 + (a + \frac{1}{\gamma})v - I_{\text{app}} = 0. \tag{6}$$

We note that by solving the cubic equation, we can obtain the analytical equilibrium points of the system. Depending on the parameters, there may be one to three solutions.

In addition, by Hartman–Grobman theorem, we can determine the stability of hyperbolic equilibrium points by studying the Jacobian of the system at each equilibrium point.

This matrix is given by the expression:

$$J(v^*) = \begin{pmatrix} P(v^*) & -1 \\ \epsilon & -\gamma\epsilon \end{pmatrix} \tag{7}$$

where $v^* = v^*(a, \gamma, I_{\text{app}})$ is a solution of (6), and $P(v) = -3v^2 + 2(a+1)v - a$. We note that stability depends on $a$, $\gamma$ and $I_{\text{app}}$, but also on $\epsilon$.

## 1.2 Bifurcation Diagram using XPPAUT

From now on, we fix the parameter values $a = 0.25$, $\gamma = 2.54$ and $\epsilon = 0.01$, and study the dynamical system in this case.

Observe that by varying the parameter $I_{\text{app}}$, the equilibrium points and their stability can change. In particular, for certain pairs $(I^*_{\text{app}}, v^*)$, the system can exhibit codimension-1 bifurcations (note that the number of independent conditions determining the bifurcation is one, since (6) must hold).

In this work, we use the software *XPPAUT* to obtain the bifurcation diagram of the system, including the stability of the equilibrium points as a function of $I_{\text{app}}$, and the emergence of stable or unstable invariant manifolds, such as limit cycles.

In order to plot the bifurcation diagram, *XPPAUT* requires an initial equilibrium point $v^*$ for a certain current value $I_{\text{app}}$. We note that by setting

$$I_{\text{app}} = 0, \tag{8}$$

the equilibrium point equation (6) can be written as:

$$v(v^2 - (a+1)v + (a + \frac{1}{\gamma})) = 0 \tag{9}$$

which yields $(v^*, w^*) = (0, 0)$ as an equilibrium point for $I_{\text{app}} = 0$. Furthermore, this is the unique equilibrium point for this current value, since the discriminant of the second order equation is negative.

We can study the stability of $(0, 0)$ by studying its Jacobian (from equation (7)):

$$J(0) = \begin{pmatrix} -a & -1 \\ \epsilon & -\gamma\epsilon \end{pmatrix} \tag{10}$$

which has eigenvalues:

$$\lambda_1 = -0.188800782772870, \quad \lambda_2 = -0.086599217227131. \tag{11}$$

Since both characteristic exponents of the equilibrium point are negative, we conclude that the equilibrium point $(0, 0)$ is an attracting node for $I_{\text{app}} = 0$.

The following figures show the bifurcation diagram obtained with *XPPAUT*. The system only exhibits one equilibrium point, which changes its stability for two different values of $I_{\text{app}}$, corresponding to subcritical Hopf bifurcations. Attracting and repelling limit cycles emerge and disappear close to the Hopf bifurcations, in saddle nodes of limit cycles.

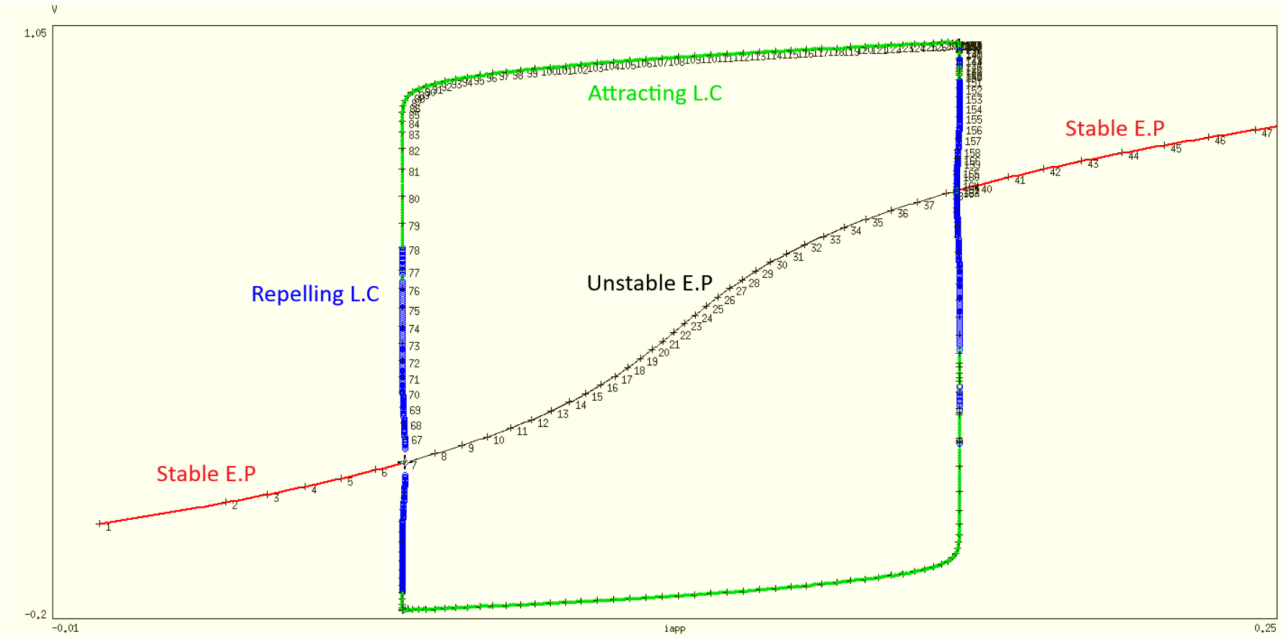


Figure 1: Bifurcation diagram obtained using *XPPAUT*. The red dots denote the values $v^*(I_{\text{app}})$ for which the equilibrium point is attracting, and the black dots those for which it is repelling. The green dots represent the upper and lower $v$ values of the attracting limit cycle as a function of $I_{\text{app}}$, and the blue dots the analogous values for the repelling limit cycle.

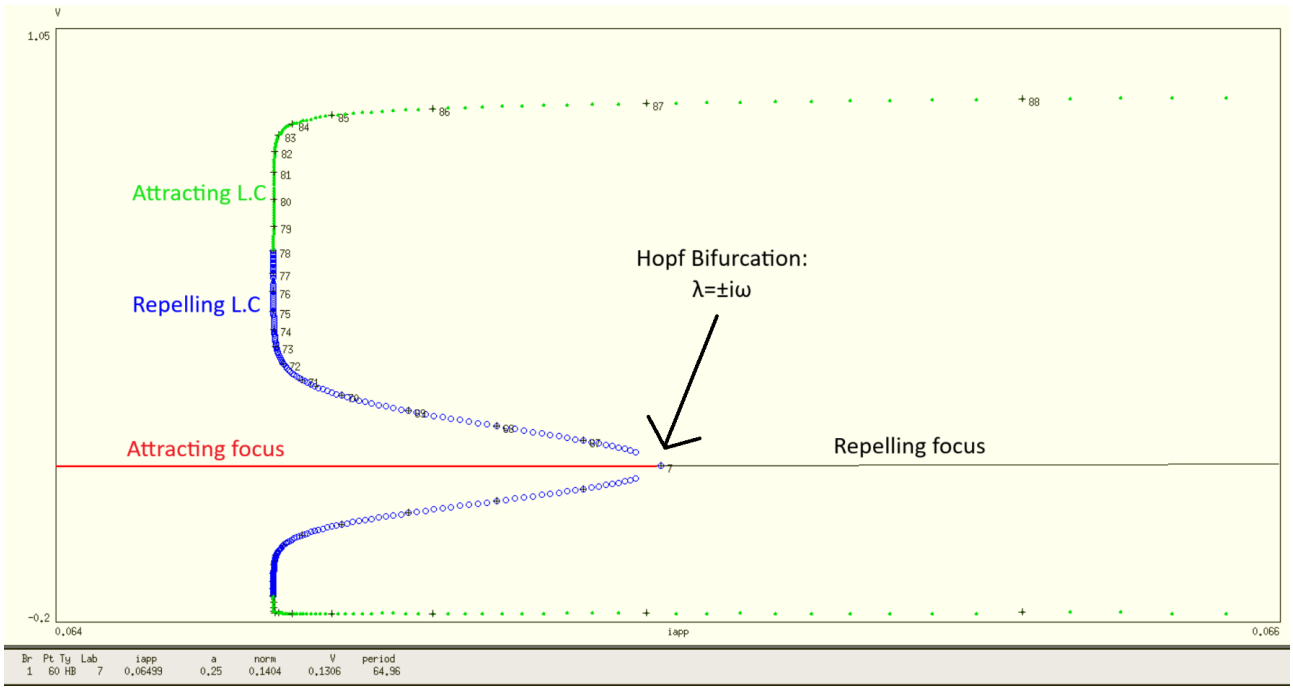


Figure 2: Bifurcation diagram obtained using *XPPAUT* near the first Hopf bifurcation, where the equilibrium point, which is a focus, transitions from stable to unstable as $I_{\text{app}}$ increases. According to *XPPAUT*, the applied current value at which the saddle node of limit cycles emerges is $I_1' = 0.06436$, while the Hopf bifurcation occurs at $I_1 = 0.06499$.

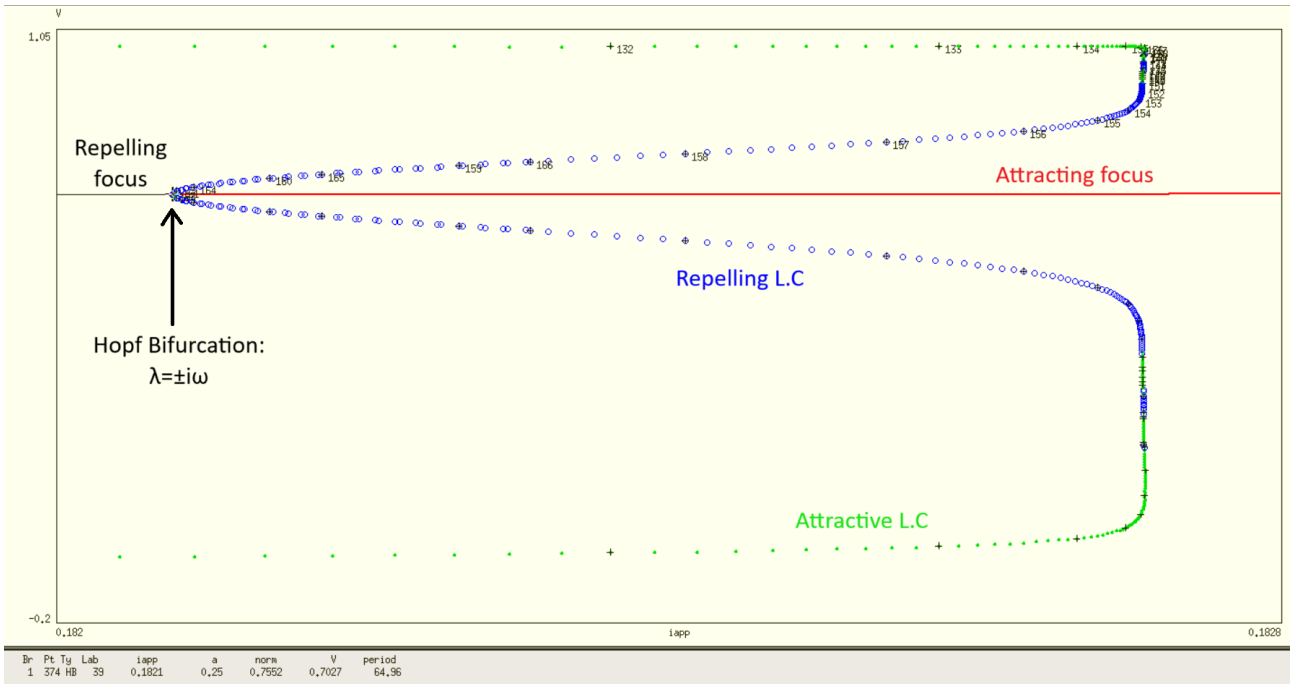


Figure 3: Bifurcation diagram obtained using *XPPAUT* near the second Hopf bifurcation, where the equilibrium point, which is a focus, transitions from unstable to stable as $I_{\text{app}}$ increases. According to *XPPAUT*, the applied current value at which the Hopf bifurcation occurs is $I_2 = 0.1821$, while the limit cycles disappear in a saddle node for $I_2' = 0.1827$.

From the bifurcation diagram, we observe that different dynamical behaviours can be distinguished in the system depending on $I_{\text{app}}$. We now present these behaviours and will analyse each region in detail in the following sections.

1. For $I_{\text{app}} \in [0, I_1')$, where $I_1' = 0.06436$, the system exhibits a single attracting equilibrium point, which acts as a global attractor.
   For these input currents, the neuronal system is **excitable**, since the asymptotic value of the membrane potential is a constant value, and increasing the current value can lead to oscillatory behaviour.

2. For $I_{\text{app}} \in (I_1', I_1)$, where $I_1' = 0.06436$ and $I_1 = 0.06499$, the system exhibits two stable structures: an attracting focus and an attracting limit cycle. It also possesses a repelling limit cycle that separates the basins of attraction of the focus and the attracting limit cycle.
   For these input currents, the neuronal system exhibits **bistability**.

3. For $I_{\text{app}} \in (I_1, I_2)$, where $I_1 = 0.06499$ and $I_2 = 0.1821$, the equilibrium point is a repeller, and the only stable structure is the attracting limit cycle. No repelling limit cycle exists in this range.
   For these input currents, the neuronal system is **oscillatory**, as the membrane potential oscillates between a maximum and a minimum value, approaching those of the attracting limit cycle.

4. For $I_{\text{app}} \in (I_2, I_2')$, where $I_2 = 0.1821$ and $I_2' = 0.1827$, the system returns to **bistability** between an attracting focus and the attracting limit cycle, and a repelling limit cycle emerges.

5. For $I_{\text{app}} \in (I_2', +\infty)$, where $I_2' = 0.1827$, there are no longer any limit cycles, and the only invariant structure is the attracting equilibrium point, which acts as a global attractor.

For these input currents, the neuronal system is in a **nerve block** state, since the asymptotic value of the membrane potential is constant, and the input current has surpassed the range in which oscillatory behaviour occurs.

The threshold values $I_{\text{app}} \in \{I_1', I_1, I_2, I_2'\}$ correspond to mathematical limits at which the system exhibits degenerate behaviour. In the following section, we describe how to analytically determine the current values at the Hopf bifurcations, $I_1$ and $I_2$.

## 1.3 Equilibrium Point Types and Hopf Bifurcations

The previous bifurcation diagram obtained using *XPPAUT* indicates that the system exhibits a change in the stability of the equilibrium point for two different values of $I_{\text{app}}$, in a Hopf bifurcation.

For a two-dimensional dynamical system depending on a parameter, in this case $I_{\text{app}}$, a Hopf bifurcation occurs when a focus changes its stability, from attracting to repelling or vice versa.

Recall that a focus is an equilibrium point whose Jacobian has a pair of complex conjugate (non-real) eigenvalues. The stability depends on the real part of these eigenvalues: the equilibrium point is an attracting focus if the real part is negative, and a repelling focus if it is positive.

Therefore, the condition that determines a Hopf bifurcation is that the real part of the complex conjugate characteristic exponents vanishes at the focus equilibrium point.

Although the equilibrium point must be a focus near the Hopf bifurcation, we observe that for $I_{\text{app}} = 0$ the equilibrium point is not a focus, but an attracting node. Indeed, the bifurcation diagram from *XPPAUT* indicates that there is a change in the stability of the equilibrium point, but does not specify if the equilibrium point is a node or a focus for each $I_{\text{app}}$. In fact, the equilibrium points transition between a node and a focus in the regions where they are stable or unstable, but this does not qualitatively affect the behaviour of the system. In the *Sagemath* code provided in the Appendix, we include a numerical computation of the stability of the equilibrium points for different values of the current, as a verification of this fact.

Let us study the Jacobian of the system evaluated at the equilibrium points, $J(v^*(I_{\text{app}}))$, to determine the equilibrium point types and the Hopf bifurcations.

From the general expression of the Jacobian (7), which has a trace and a determinant

$$Tr(J(v^*)) = P(v^*) - \gamma\epsilon, \tag{12}$$

$$det(J(v^*)) = \epsilon(1 - \gamma P(v^*)), \tag{13}$$

we obtain a characteristic polynomial

$$\lambda^2 + (\gamma\epsilon - P(v^*))\lambda + \epsilon(1 - \gamma P(v^*)). \tag{14}$$

The roots of the characteristic polynomial are the eigenvalues

$$\lambda_\pm = \frac{1}{2}(P(v^*) - \gamma\epsilon) \pm \frac{i}{2}\sqrt{Q(v^*)}, \tag{15}$$

where

$$Q(v) = 4\epsilon(1 - \gamma P(v)) - (P(v) - \gamma\epsilon)^2. \tag{16}$$

From the previous expression, we deduce that the equilibrium point $(v^*, w^*)$ will be a focus provided that

$$Q(v^*) > 0. \tag{17}$$

If the previous condition is not satisfied, the equilibrium point is either an attracting node or a repelling node, as indicated by the *XPPAUT* bifurcation diagram.

Now, let us suppose that the focus condition (17) is satisfied. Then, the Hopf bifurcation condition can be expressed as

$$P(v^*) - \gamma\epsilon = -3v^{*2} + 2(a+1)v^* - (a + \gamma\epsilon) = 0 \tag{18}$$

which describes the possible membrane potential values $v^*$ at which the Hopf bifurcation occurs.

The roots of the previous second order equation can be expressed as:

$$v^*_{1,2} = \frac{a+1}{3}(1 \pm \Delta), \tag{19}$$

where

$$\Delta^2 = 1 - 3\frac{a + \gamma\epsilon}{(a+1)^2}. \tag{20}$$

The value $\Delta^2$ is positive for the chosen parameter values, which implies that expression (19) yields two possible membrane potentials, $v^*_{1,2}$. Since these potentials also satisfy the focus condition (17), we conclude that Hopf bifurcations occur for these membrane potential values.

Finally, the values of $I_{\text{app}}$ at which the Hopf bifurcations occur are obtained from relation (6), which is satisfied at any equilibrium point. Hence, we obtain the current values:

$$I_{1,2} = v^{*3}_{1,2} - (a+1)v^{*2}_{1,2} + (a + \frac{1}{\gamma})v^*_{1,2}. \tag{21}$$

The numerical values of the Hopf bifurcations, $(v^*_{1,2}, w^*_{1,2})$ and $I_{1,2}$, can be obtained by directly substituting the previous expressions (19) and (21). These computations are performed in the *Sagemath* code, where we also verify that the focus condition (17) is satisfied and that (20) takes a positive value.

The obtained numerical values are the following:

$$I_1 = 0.064989258896071 \tag{22}$$
$$v^*_1 = 0.130640249860262 \tag{23}$$
$$w^*_1 = 0.051433169236324 \tag{24}$$

$$I_2 = 0.182076212086723 \tag{25}$$
$$v^*_2 = 0.702693083473071 \tag{26}$$
$$w^*_2 = 0.276650820264989 \tag{27}$$

which are in agreement with those obtained using the *XPPAUT* software.

## 1.4 Phase Portraits for Different Regimes

In this section, we use *XPPAUT* to obtain the phase portraits of the system for input current values $I_{\text{app}}$ corresponding to the different regimes.

### 1.4.1 Phase Portrait in the Excitable Regime

If we fix an input current $I_{\text{app}} = 0.05 \in [0, I_1')$, the system is in the excitable regime. The equilibrium point is a global attractor, which implies that the $\omega$-limit of any initial condition $(v_0, w_0)$ is the equilibrium point $(v^*, w^*)$.

The phase portrait of the system in the nerve block regime is qualitatively the same as in the excitable regime.

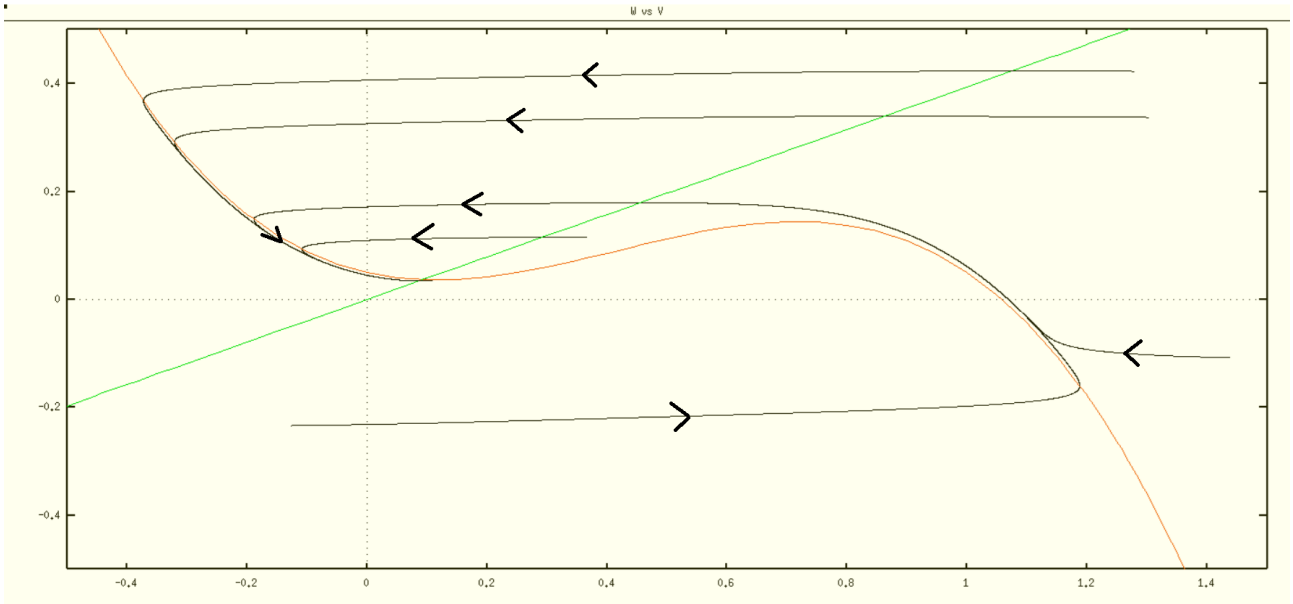


Figure 4: Phase portrait of the system for $I_{\text{app}} = 0.05$, in the excitable regime. The green line represents the $w$-nullcline, and the orange curve the $v$-nullcline. Some representative trajectories are shown, all of which have as their $\omega$-limit the attractive equilibrium point.

In this first phase portrait, we clearly see that the trajectories of the system are almost horizontal until they reach the $v$-nullcline. Then, they follow the $v$-nullcline until they reach the equilibrium point.

This behaviour is characteristic of a slow–fast dynamical system, which has an associated vector field oriented along the direction of the fast variable (in this case, the horizontal one, $v$), except in a neighbourhood of the fast variable's nullcline, where the vector field is oriented along the direction of the slow variable (in this case, the vertical one, $w$).

### 1.4.2 Phase Portrait in the Bistability Regime

If we fix an input current $I_{\text{app}} = 0.0648 \in (I_1', I_1)$, the system is in the bistability regime. In this case, the equilibrium point is an attracting focus, behaving almost as a centre since we are very close to the Hopf bifurcation. In this regime, there also exist a repelling and an attracting limit cycle.

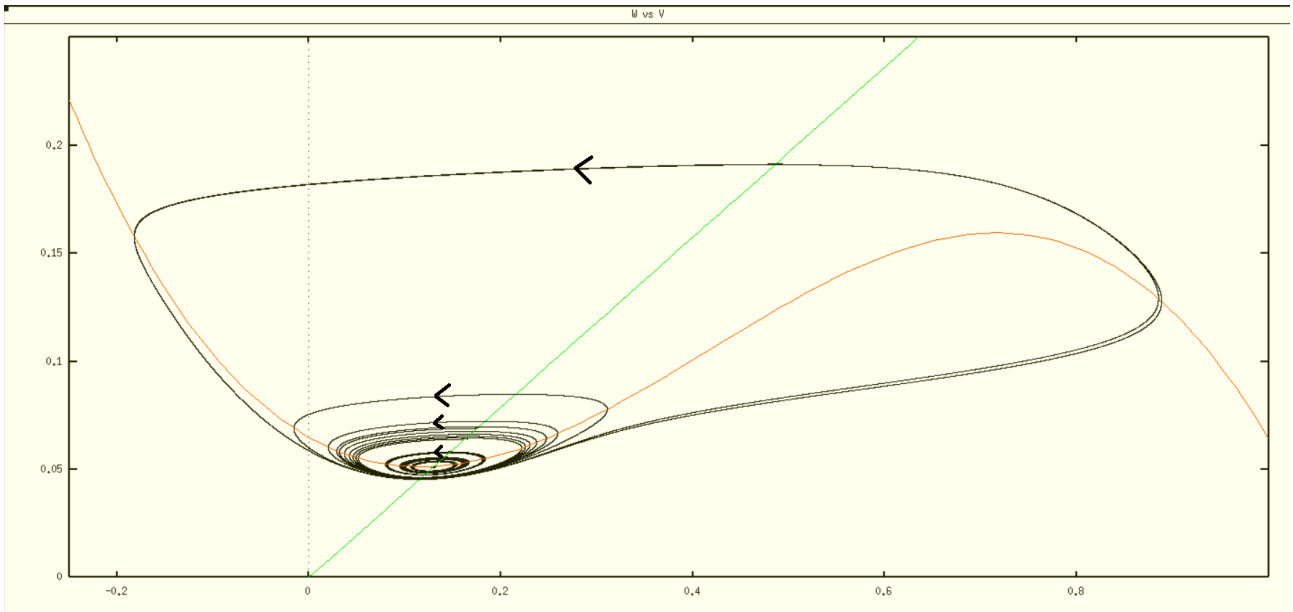


Figure 5: Phase portrait of the system for $I_{\text{app}} = 0.0648$, in the bistability regime. The green line represents the $w$-nullcline, and the orange curve the $v$-nullcline. Some representative trajectories are shown, all of which have as their $\omega$-limit either the attracting focus or the attracting limit cycle.

From the phase portrait, we observe two different types of $\omega$-limits:

1. The attracting focus, which behaves almost as a centre. Its basin of attraction is the interior of the bounded region whose boundary is the repelling limit cycle. Any orbit starting from a point (different from the equilibrium point) within this basin will behave almost periodically, approaching asymptotically the equilibrium point.

2. The attracting limit cycle, whose basin of attraction consists of the remaining points in the plane.

### 1.4.3 Phase Portrait in the Oscillatory Regime

If we fix an input current $I_{\text{app}} = 0.12 \in (I_1, I_2)$, the system is in the oscillatory regime. The equilibrium point is a repeller, and there exists a single attracting limit cycle. The $\omega$-limit of any initial condition $(v_0, w_0)$ is this attracting limit cycle. The time required for the variables to reach the oscillatory behaviour described by the limit cycle is on the order of its period.

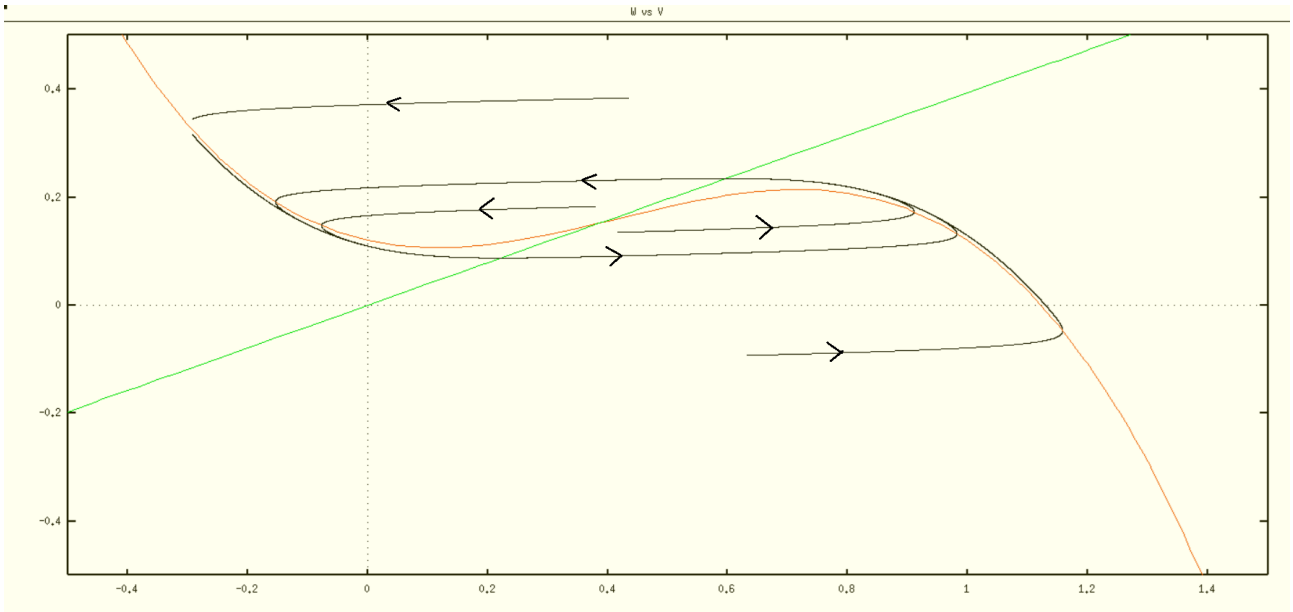


Figure 6: Phase portrait of the system for $I_{\text{app}} = 0.12$, in the oscillatory regime. The green line represents the $w$-nullcline, and the orange curve the $v$-nullcline. Some representative trajectories are shown, all of which have as their $\omega$-limit the attracting limit cycle.

## 1.5 Voltage Time Evolution in the Bistability Region

In the bistability regimes $(I_1', I_1)$ and $(I_2, I_2')$, there are two different attractors: the attracting focus and the attracting limit cycle, as previously discussed and illustrated in the corresponding phase portrait.

Here, we show the time evolution of the membrane potential, $v(t)$, for initial conditions $(v_0, w_0)$ within the basins of attraction of the different attractors, using *XPPAUT*.

Specifically, for $I_{\text{app}} = 0.0644 \in (I_1', I_1)$, we consider the following initial conditions:

$$(v_0, w_0) = (0.13, 0.05) \tag{28}$$

which has the attracting focus as its $\omega$-limit, and

$$(v_0, w_0) = (0.26, 0.05), \tag{29}$$

which has the attracting limit cycle as its $\omega$-limit.

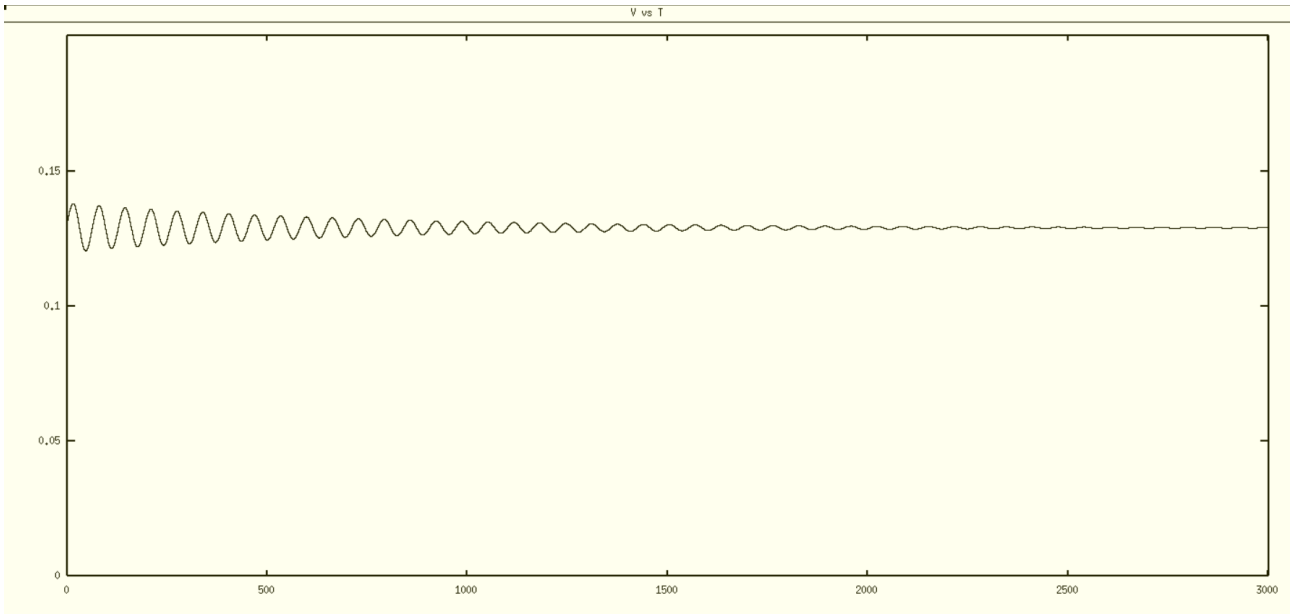


Figure 7: Plot of the time evolution of the membrane potential, $v(t)$, for $I_{\text{app}} = 0.064$ and $(v_0, w_0) = (0.13, 0.05)$. The voltage oscillates around the value $v^*(I_{\text{app}})$, to which it tends asymptotically.

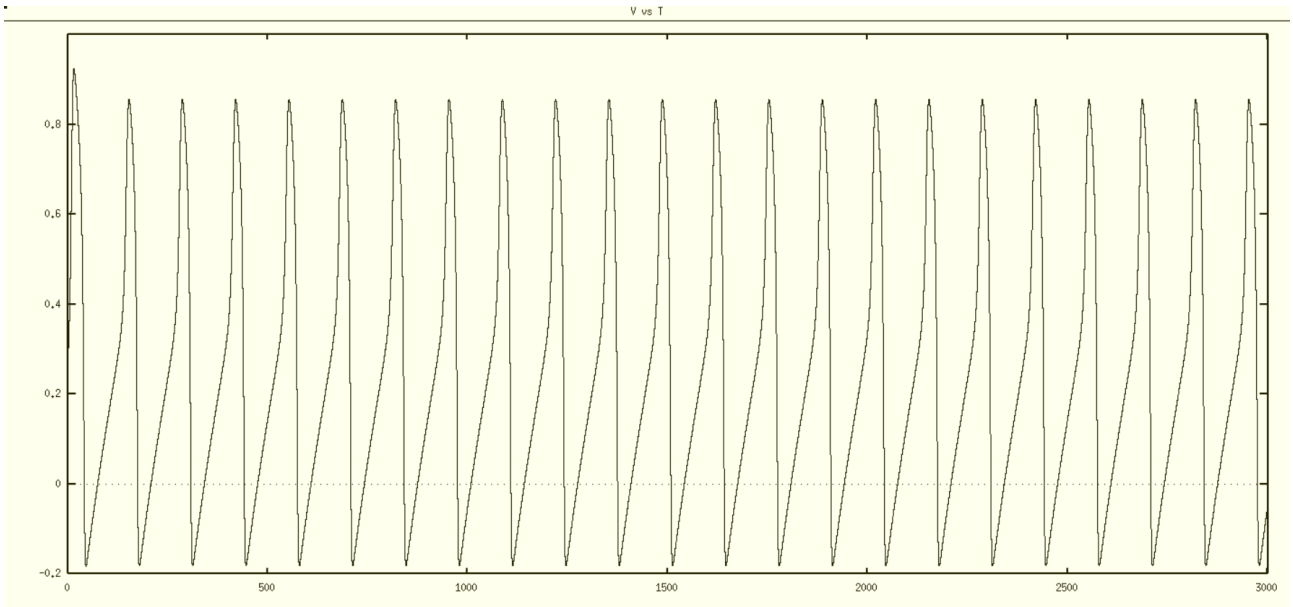


Figure 8: Plot of the time evolution of the membrane potential, $v(t)$, for $I_{\text{app}} = 0.064$ and $(v_0, w_0) = (0.26, 0.05)$. The voltage rapidly evolves towards a periodic oscillation, with maximum and minimum values corresponding to those of the attracting limit cycle.

## 1.6 Frequency of Oscillations in the Oscillatory Regime

In the oscillatory regime, $I_{\text{app}} \in (I_1, I_2)$, the orbits rapidly approach the attracting limit cycle, where the variables $v$ and $w$ oscillate between well-defined maximum and minimum values.

In this section, we plot the frequency of these oscillations as a function of the input current $I_{\text{app}}$ using *XPPAUT*.

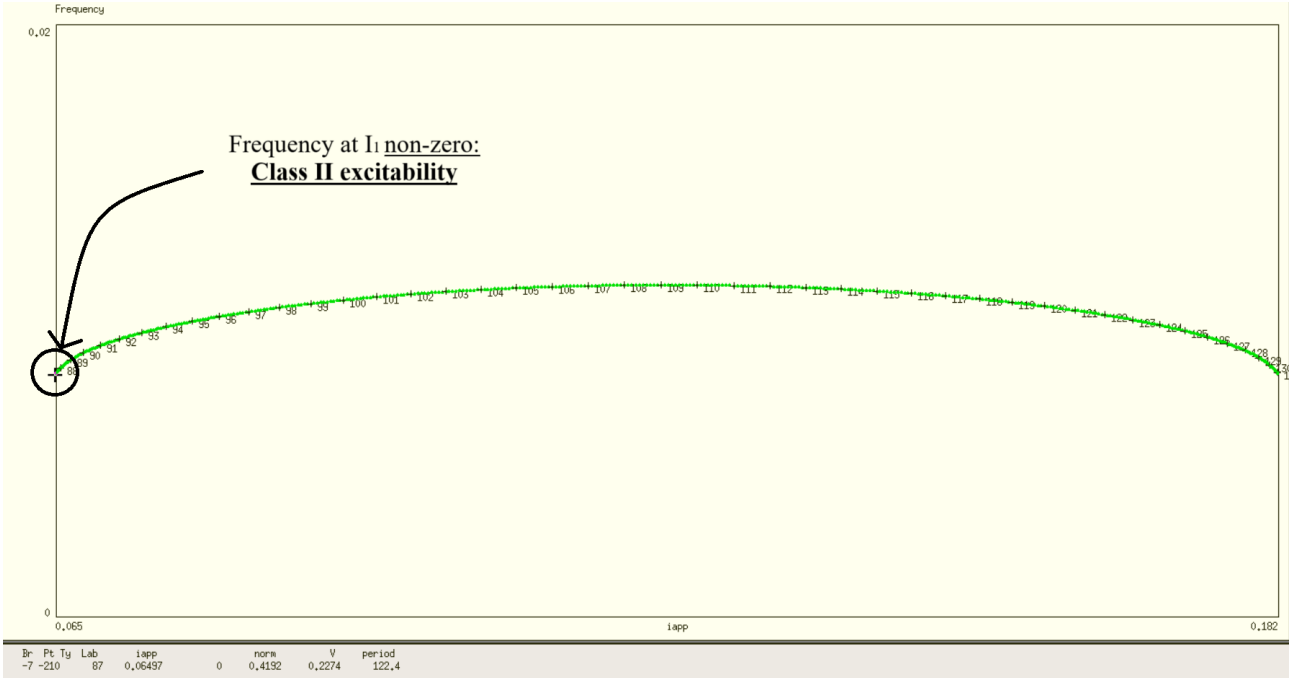


Figure 9: Plot of the frequency of the system's oscillations in the oscillatory regime as a function of $I_{\text{app}}$.

We note that at the threshold value $I_{\text{app}} = I_1$, the oscillation frequency takes a non-zero value; in fact, the attracting limit cycle emerges in the bistability region with a non-zero frequency.

Hence, in the oscillatory regime, the considered model exhibits **class II excitability**.

## 1.7 Computation of the Period in the Oscillatory Regime

We have commented that the system exhibits slow-fast dynamics, such that the periodic orbits of the system in the oscillatory regime spend much of the time on the $v$-nullcline ($\dot{v} = 0$), which behaves as a slow manifold. Then, the period of the orbit can be computed as

$$T = \frac{1}{\varepsilon} \int_{W_-}^{W_+} \left( \frac{1}{G_+(w) - \gamma w} - \frac{1}{G_-(w) - \gamma w} \right) dw,$$

where $G_-(w)$ and $G_+(w)$ are the expressions of the two slow branches of the cubic- shaped nullcline $\dot{v} = 0$, and $W_-$ (resp., $W_+$) is the local minimum (resp., maximum) of this cubic.

Let us see why this is true:

### 1.7.1 The Nullclines

The $v$–nullcline is given by:

$$0 = -v(v-1)(v-a) - w + I_{\text{app}}, \tag{30}$$

which can be written as:

$$w = F(v) := -v(v-1)(v-a) + I_{\text{app}}. \tag{31}$$

This is a cubic function in $v$, with two turning points corresponding to the local minimum and maximum at:

$$(W_-, W_+) = \big(F(v_{\min}), F(v_{\max})\big).$$

For each $w \in [W_-, W_+]$, there are three possible $v$ values satisfying $w = F(v)$. The two monotonic (stable) outer branches can be written as inverse functions:

$$v = G_-(w) \quad \text{(left branch)}, \qquad v = G_+(w) \quad \text{(right branch)}.$$

### 1.7.2 Slow Dynamics on Each Branch

On the slow manifold (i.e., along the nullcline $\dot{v} = 0$), we have $v \approx G_\pm(w)$. Substituting this into Eq. (2) gives the reduced slow dynamics:

$$\dot{w} = \varepsilon\big(G_\pm(w) - \gamma w\big). \tag{32}$$

By definition, $\dot{w} = \dfrac{dw}{dt}$, so we can write

$$dt = \frac{dw}{\dot{w}} = \frac{dw}{\varepsilon\big(G_\pm(w) - \gamma w\big)}.$$

Integrating both sides between two points $w_1$ and $w_2$ along the same slow branch yields the total time spent moving along that branch:

$$\Delta t = \int_{t_1}^{t_2} dt = \int_{w_1}^{w_2} \frac{dw}{\varepsilon\big(G_\pm(w) - \gamma w\big)}. \tag{33}$$

### 1.7.3 Total Period of the Relaxation Oscillation

The relaxation oscillation spends most of its time on the two slow branches. The fast jumps between them occur on a timescale $\mathcal{O}(1)$, negligible compared to $\mathcal{O}(1/\varepsilon)$.

- On the **right branch** ($v = G_+(w)$), the system evolves from $w = W_-$ to $w = W_+$:

$$T_+ = \frac{1}{\varepsilon}\int_{W_-}^{W_+} \frac{dw}{G_+(w) - \gamma w}.$$

- On the **left branch** ($v = G_-(w)$), it evolves back from $w = W_+$ to $w = W_-$:

$$T_- = \frac{1}{\varepsilon}\int_{W_+}^{W_-} \frac{dw}{G_-(w) - \gamma w} = -\frac{1}{\varepsilon}\int_{W_-}^{W_+} \frac{dw}{G_-(w) - \gamma w}.$$

The total period is therefore

$$\begin{aligned} T &= T_+ + T_- \\ &= \frac{1}{\varepsilon}\int_{W_-}^{W_+} \left[\frac{1}{G_+(w) - \gamma w} - \frac{1}{G_-(w) - \gamma w}\right] dw. \end{aligned} \tag{34}$$

### 1.7.4 Final Expression

Equation (34) gives the leading-order approximation for the period of the relaxation oscillation in the FitzHugh–Nagumo model:

$$T = \frac{1}{\varepsilon} \int_{W_-}^{W_+} \left( \frac{1}{G_+(w) - \gamma w} - \frac{1}{G_-(w) - \gamma w} \right) dw. \tag{35}$$

We note that this approximation becomes more accurate for smaller values of $\epsilon$, since as $\epsilon \to 0$, the system spends a longer time on the slow manifold compared to the fast regime. Therefore, for smaller values of $\epsilon$, the period $T$ will increase (the branches $G_\pm$ do not depend on $\epsilon$) and the approximation will be better.

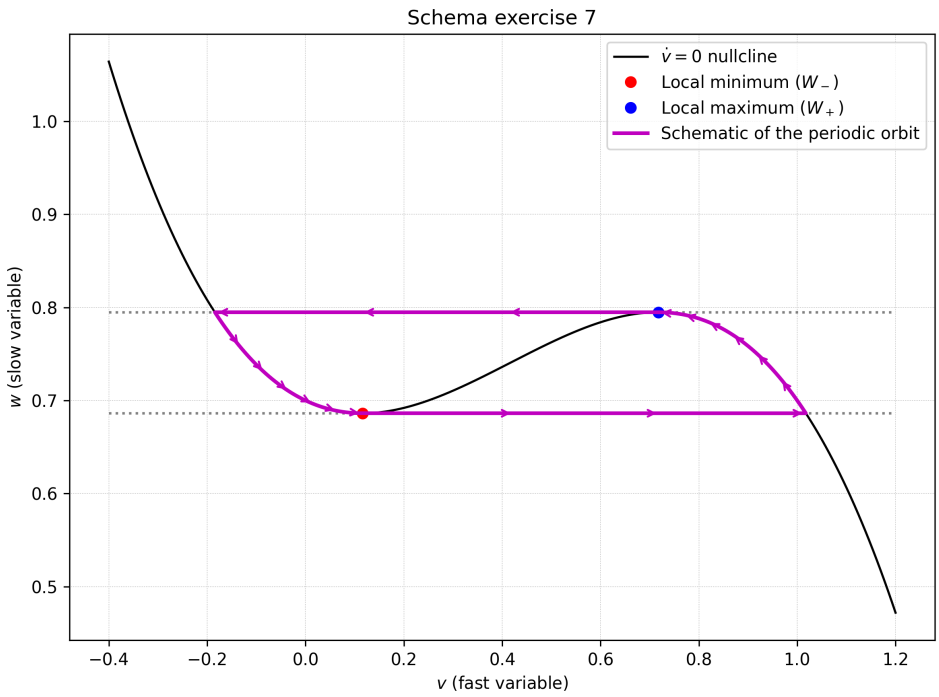


Figure 10: Phase-plane diagram of the FitzHugh–Nagumo model showing the cubic-shaped $v$-nullcline and the relaxation oscillation. The points $W_-$ and $W_+$ correspond to the local minimum and maximum of the cubic nullcline, respectively. The trajectory follows the slow manifolds (colored branches) and transitions rapidly between them near $W_\pm$ due to the fast–slow dynamics.

# 2 BURSTING

## 2.1 The Model

Bursting systems are systems characterized by periods of spiking (bursts) followed by periods of resting states. An example of a bursting system is given by the following set of differential equations:

$$\begin{cases} C\dot{V} = I - g_L(V - E_L) - g_{Na}\, a\, m_\infty(V)\,(V - E_{Na}) \\ \qquad\qquad - g_K\, n\,(V - E_K) - g_w\, w\,(V - E_K), \\ \tau_n \dot{n} = n_\infty(V) - n, \\ \tau_w \dot{w} = w_\infty(V) - w \end{cases} \tag{36}$$

where

$$m_\infty(V) = \frac{1}{1 + \exp\left(\frac{v_m - V}{k_m}\right)}, \quad n_\infty(V) = \frac{1}{1 + \exp\left(\frac{v_n - V}{k_n}\right)}, \quad w_\infty(V) = \frac{1}{1 + \exp\left(\frac{v_w - V}{k_w}\right)}.$$

We will use the following parameters:

$$\begin{aligned} &C = 1\ \mu\text{F}, && E_L = -80\ \text{mV}, && E_{\text{Na}} = 60\ \text{mV}, && E_K = -90\ \text{mV}, \\ &g_L = 8\ \text{mS}, && g_{\text{Na}} = 20\ \text{mS}, && g_K = 9\ \text{mS}, && g_w = 5\ \text{mS}, \\ &v_m = -20, && k_m = 15, && v_n = -25, && k_n = 5, \\ &\tau_n = 0.154\ \text{ms}, && v_w = -20, && k_w = 5, && \tau_w = 20\ \text{ms}, \\ &I = 5. \end{aligned}$$

We identify a *fast* subsystem, constituted by the first two equations of (36), which is the system associated to spike generation. The third equation describes the time evolution of the *slow* variable, since $\tau_w >> \tau_n$.

## 2.2 Identifying the Slow-Fast Subsystems

If one looks at the plots of $V$, $n$ and $w$ with respect to time (Figures 11, 12 and 13 respectively), one can clearly see that the fast variable is $V$ and that $w$ is the slow variable: $V$ presents spikes (rapid and big changes) during the bursting period, while $w$ doesn't present this fast changing behavior, and instead presents a very slow recovery state. This is why we can consider $w$ as the variable that "governs" the change between bursting periods and resting periods. The variable $n$ lies in between $V$ and $w$: while it doesn't present such big changes as $V$, it also presents spikes during the bursting period and a fast recovery to the resting state. This is why the system $(V, n)$ is called the fast system.

## 2.3 Bifurcation Diagram

Taking this into consideration, as $w$ is the slow variable, we can consider just the fast system and take $w$ as a parameter. Doing so enables us to plot a bifurcation diagram (Figure 14) of the fast system, where we can see 2 different bifurcations: a Hopf bifurcation at point 2 of the plot (subcritical), and a saddle-node bifurcation at point 5 of the plot. We will be mostly interested in the saddle-node bifurcation for the dynamics of the bursting.

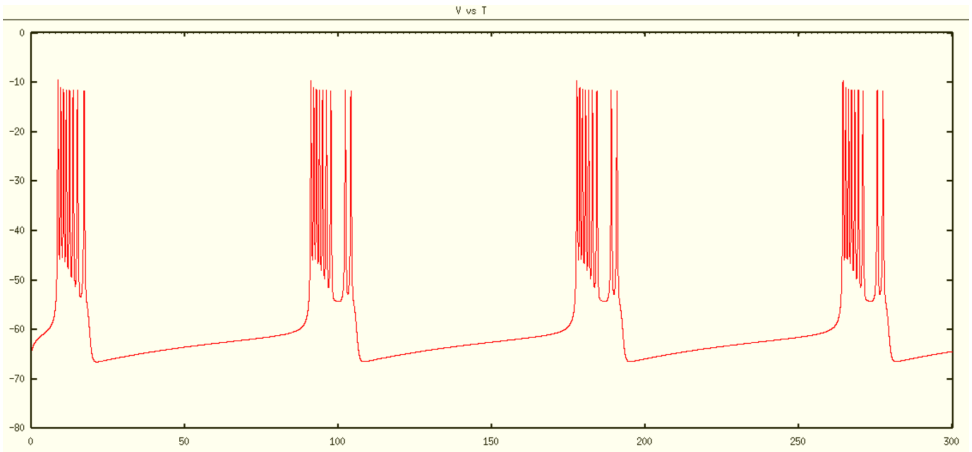


Figure 11: Plot of the fast variable $V$ with respect to time.

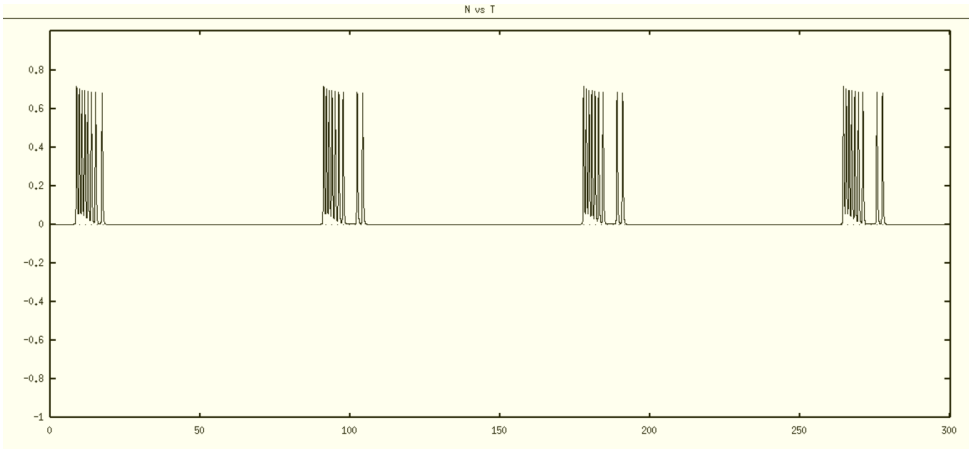


Figure 12: Plot of the variable $n$ with respect to time.

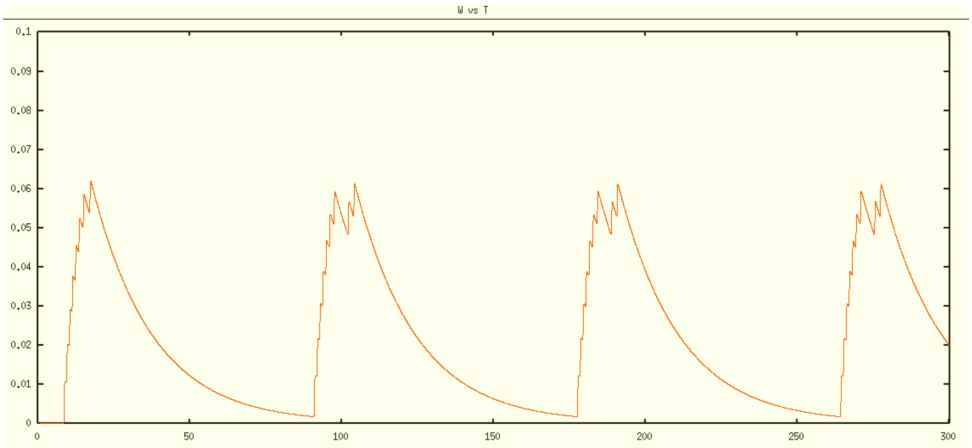


Figure 13: Plot of the slow variable $w$ with respect to time.

In fact, there is also a homoclinic bifurcation: the stable and unstable manifolds of the saddle coincide and form a homoclinic orbit. This can be seen in Figure 15, where we overlap the bifurcation diagram of the fast system with the bursting dynamics of the whole system (i.e., also taking $w$ as a variable) projected onto the $w$ - $V$ axis. What we can see is that the curve starting at the Hopf bifurcation goes very far away from the unstable point and oscillates until it reaches the lower unstable branch

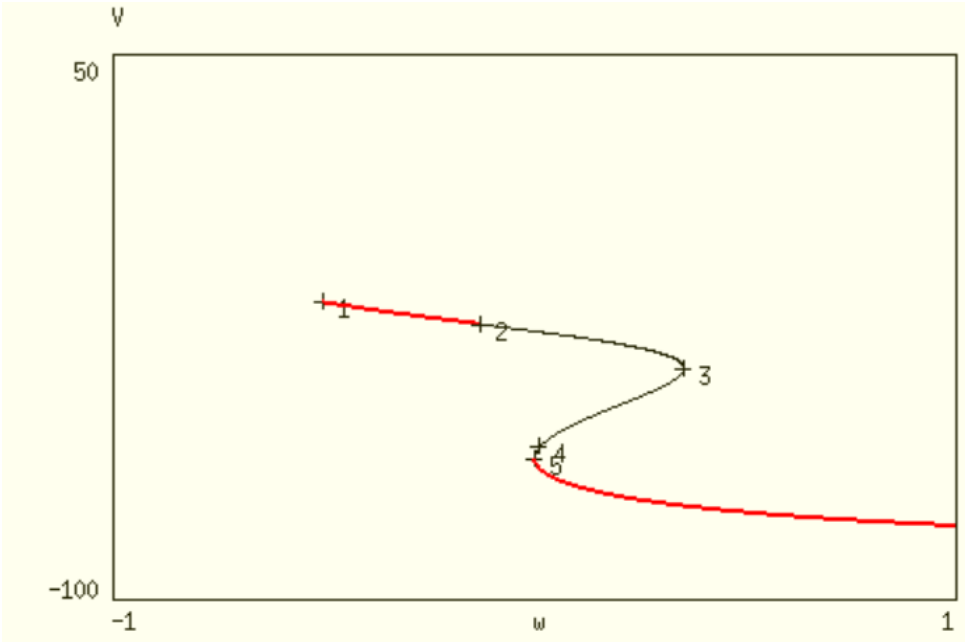


Figure 14: Bifurcation diagram considering the fast system $(V, n)$ and taking $w$ as a parameter. Red points indicate attracting behavior, and black points instability (in this case, a saddle point).

and crosses it, indicating the presence of a homoclinic bifurcation, as it approaches the saddle, but after the bifurcation, the orbit goes to the stable node (lowest branch in red). The bursting process is explained through this:

First, there are oscillations (spikes) for increasing values of $w$ which start to tend to the homoclinic orbit, until they cross the unstable lower branch (in black) after the homoclinic bifurcation, and thus then tend to the stable node (red lower branch). Then, as we are approaching a stable equilibrium point, the dynamics are very slow and $w$ decreases slowly (leading to the slow recovery rate of the variable $w$ after the bursting process). This behavior continues on until the saddle-node bifurcation is reached, when the stable equilibrium node disappears and the system bounces back up again and starts oscillating (generating spikes).

Another thing we can observe is that, when the "burst" (or period of spiking) is getting near the end, as the system is tending to a homoclinic orbit, the time between two consecutive spikes slows down, as seen in Figure 11. This is because a homolcinic orbit ends at an equilibrium point, and thus the dynamics slow down (it takes an infinite amount of time for a homoclinic orbit to reach the equilibrium point it tends to).

With all this, we can conclude that the system is a square-wave bursting system.

## 2.4 Other Models of Bursting

Over many more, there are two other bursting system that we will explain: parabolic bursting and elliptic bursting.

In parabollic bursting (or circle/circle bursting), the bursting starts after a resting state due to a saddle-node on invariant circle bifurcation, and the bursting state (or spiking) is halted when reaching another saddle-node on invariant circle bifurcation. In these systems, there is usually no hysteresis and the system has at least two slow variables (although this is not always the case).

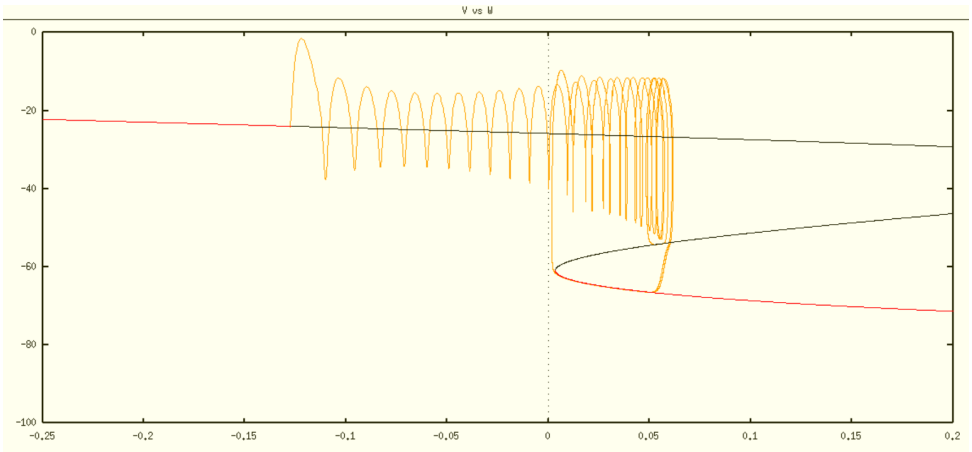


Figure 15: Overlap of the bifurcation diagram of the fast system with the projection of the bursting trajectory for the whole system in the $w$ - $V$ axis.

On the other hand, in elliptic bursting (or SubHopf/Fold cycle), the bursting starts after a resting state due to a subcritical Hopf bifurcation, and the bursting state (or spiking) is halted when reaching a fold limit cycle bifurcation. In these systems, there is usually hysteresis and the system has only one slow variable.

# 3 APPENDIX: SAGEMATH CODE

```
#! /usr/bin/env sage
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.colors import Normalize
from scipy.integrate import solve_ivp



#Fitz Hugh-Nagumo 2D model
def f(v,w,a,I_app,epsilon,gamma):
    dvdt=-v*(v-1)*(v-a)-w+I_app
    dwdt=epsilon*(v-gamma*w)
    return [dvdt,dwdt]

def Df(v,w,a,I_app,epsilon,gamma): #Although does not depend explicitely on I_app
    return [[-3*v*v+2*(a+1)*v-a,-1],[epsilon,-epsilon*gamma]]

def ep(a,gamma,I_app):
    #Define polynomial (SageMath)
    R.<x>=PolynomialRing(RR)
    p=x^3-(a+1)*x^2+(a+1/gamma)*x-I_app

    #Exact roots (valid for degree  5)
    exact_roots=p.roots(multiplicities=False)
    epts=[]
    for v in exact_roots:
        epts.append([v,v/gamma])
    return epts

def eigen_print(a,b,c,d):
    A=np.array([[a,b],[c,d]])
    print("Matrix A:")
    print(A)
    evals,_=np.linalg.eig(A)
    print("\nEigenvalues:")
    for i, in enumerate(evals):
        print(f"{i+1} = {:.16f}")
    print("\n")

def eigen(a,b,c,d):
    A=np.array([[a,b],[c,d]])
    evals,_=np.linalg.eig(A)
    return evals
```

```
#Hopf bifurcation functions

def Delta(a,epsilon,gamma):
    x=1-3*(a+epsilon*gamma)/(a+1)**2
    if x<0:
        raise ValueError("Hopf Bifurcations do not exist (A2 not fullfiled)")
    else:
        return sqrt(x)

def v_ep_HB(a,Delta):
    return [(a+1)*(1+Delta)/3,(a+1)*(1-Delta)/3]

def P(v,a):
    return -3*v*v+2*(a+1)*v-a

def K(p,epsilon,gamma):
    return 4*epsilon*(1-gamma*p)-(p-epsilon*gamma)**2


def HB(a,epsilon,gamma):
    D=Delta(a,epsilon,gamma)
    v_ep=v_ep_HB(a,D)
    v_ep_valid=[]
    for v in v_ep:
        p=P(v,a)
        k=K(p,epsilon,gamma)
        if k<0 or k==0:
            print(f"HB does not exist for {v:14f} (A1 not fullfiled)")
        else:
            v_ep_valid.append(v)

    if len(v_ep_valid)==0:
        print("Hopf Bifurcations do not exist (A1 never fullfiled)")
    else:
        HB_points=[]
        for v in v_ep_valid:
            I_app=v**3-(a+1)*v*v+(a+1/gamma)*v
            HB_points.append([I_app,[v,v/gamma]])
        return HB_points



#MAIN:

epsilon=0.01
gamma=2.54
a=0.25
```

```
#Stability of a certain fixed point
I_app=0
epts=ep(a,gamma,I_app)
for p in epts:
    A=Df(p[0],p[1],a,I_app,epsilon,gamma)
    eigen_print(A[0][0],A[0][1],A[1][0],A[1][1])


print("\n")

#Stability of the fixed points for different I_values:
I_values=np.linspace(0,0.25,101)
stab=[]
print("I_app:")
print("evals:")
for I_app in I_values:
    epts=ep(a,gamma,I_app)
    for p in epts:
        A=Df(p[0],p[1],a,I_app,epsilon,gamma)
        evals=eigen(A[0][0],A[0][1],A[1][0],A[1][1])
        stab.append([I_app,evals])

for x in stab:
    print(x[0])
    print(x[1])
    print("\n")
print("\n")


#Hopf bifurcations:
HB_points=HB(a,epsilon,gamma)
for p in HB_points:
    print("HB at [I_app,[v*,w*]]=")
    print(p)
    print("\n")
    print("Comprovation:")
    print("\n")

    I_app=p[0]
    v=p[1][0]
    w=p[1][1]


    A=Df(v,w,a,I_app,epsilon,gamma)
    ax=A[0][0]
    bx=A[0][1]
    cx=A[1][0]
    dx=A[1][1]
```

```
eigen_print(ax,bx,cx,dx)
```